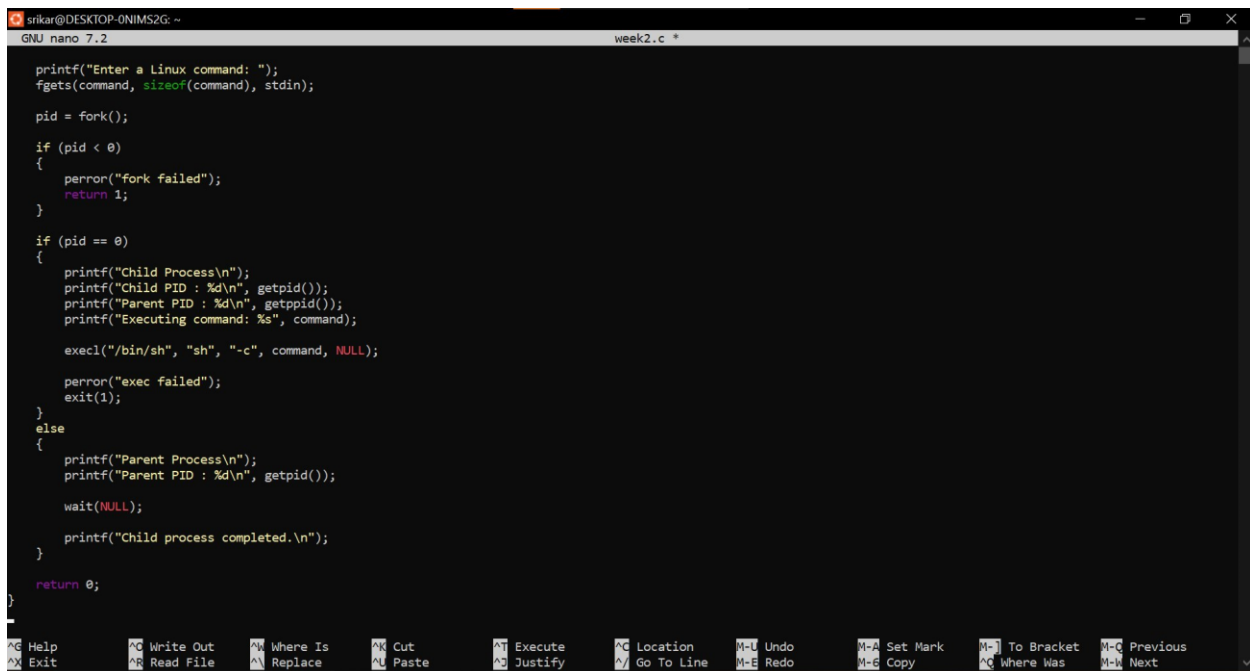


WEEK 2

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using fork() system call.
- b) Read the user input as shell command.
- c) Use a system call exec() to execute the shell command.
- d) Use a wait() system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.



```
printf("Enter a Linux command: ");
fgets(command, sizeof(command), stdin);

pid = fork();

if (pid < 0)
{
    perror("fork failed");
    return 1;
}

if (pid == 0)
{
    printf("Child Process\n");
    printf("Child PID : %d\n", getpid());
    printf("Parent PID : %d\n", getppid());
    printf("Executing command: %s", command);

    execl("/bin/sh", "sh", "-c", command, NULL);

    perror("exec failed");
    exit(1);
}
else
{
    printf("Parent Process\n");
    printf("Parent PID : %d\n", getpid());

    wait(NULL);

    printf("Child process completed.\n");
}

return 0;
```

GNU nano 7.2 week2.c *

^G Help ^G Write Out ^X Where Is ^K Cut ^T Execute ^C Location ^U Undo ^M-A Set Mark ^M-T To Bracket ^M-Q Previous
^X Exit ^R Read File ^W Replace ^L Paste ^J Justify ^_ Go To Line ^M-E Redo ^M-G Copy ^M-W Where Was ^M-N Next

```

Parent PID : 1981
Executing command: 1s
sh: 1: 1s: not found
Child process completed.
srikar@DESKTOP-0NIMS2G:~$ ./week2
Enter a Linux command: ls
Parent Process
Parent PID : 2632
Child Process
Child PID : 2804
Parent PID : 2632
Executing command: ls
File1.txt      file.c      s3p1      s3t2      skillweek-4  week2.c
File2.txt      hello.py.save s3p1.c    s3t2.c    task1.c      week_5.c
OSSP           myshell     s3p2      s3t3      task1.c.save
Skill_week3    practical.c s3p2.c    s3t3.c    week2
Child process completed.
srikar@DESKTOP-0NIMS2G:~$ █

```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```

srikar@DESKTOP-0NIMS2G: ~
srikar@DESKTOP-0NIMS2G:~$ nano week2.c
srikar@DESKTOP-0NIMS2G:~$

```

```
srikar@DESKTOP-ONIMS2G: ~
top - 05:01:26 up 1:24, 1 user, load average: 0.12, 0.12, 0.06
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 99.6 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7801.6 total, 7238.7 free, 528.6 used, 184.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7273.1 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0 21672 13144 9784 S   0.7   0.2   0:08.62 systemd
    2 root        20   0 3180  2212 2072 S   0.0   0.0   0:00.03 init-systemd(Ub
    7 root        20   0 3196  2132 2008 S   0.0   0.0   0:00.00 init
   53 root       19  -1 50460 18584 17436 S   0.0   0.2   0:01.98 systemd-journal
  101 root       20   0 25136  6576 5172 S   0.0   0.1   0:00.49 systemd-udev
  162 systemd+  20   0 21468 13192 10932 S   0.0   0.2   0:00.10 systemd-resolve
  163 systemd+  20   0 91036  7984 7016 S   0.0   0.1   0:00.07 systemd-timesyn
  172 root       20   0 4248  2680 2428 S   0.0   0.0   0:00.01 cron
  173 message+  20   0 9600  5352 4676 S   0.0   0.1   0:01.38 dbus-daemon
  180 root       20   0 17976  8820 7780 S   0.0   0.1   0:00.54 systemd-logind
  198 syslog    20   0 222520 5660 4580 S   0.0   0.1   0:00.76 rsyslogd
  200 root       20   0 3128  1972 1832 S   0.0   0.0   0:00.01agetty
  210 root       20   0 112804 25092 15144 S   0.0   0.3   0:00.15 unattended-upgr
  300 root       20   0 6700  4648 3872 S   0.0   0.1   0:00.01 login
  345 srikar     20   0 20128 11212 9268 S   0.0   0.1   0:00.13 systemd
  347 srikar     20   0 21168  3576 1848 S   0.0   0.0   0:00.00 (sd-pam)
  382 srikar     20   0 6084  5324 3660 S   0.0   0.1   0:00.02 bash
 1348 root       20   0 3184  1124  980 S   0.0   0.0   0:00.00 SessionLeader
 1349 root       20   0 3200  1260 1108 S   0.0   0.0   0:00.14 Relay(1353)
 1353 srikar     20   0 6448  5632 3652 S   0.0   0.1   0:00.08 bash
 4072 root       20   0 3184  1124  980 S   0.0   0.0   0:00.00 SessionLeader
 4073 root       20   0 3200  1260 1108 S   0.0   0.0   0:00.04 Relay(4079)
 4079 srikar     20   0 6448  5652 3672 S   0.0   0.1   0:00.04 bash
 4790 root       20   0 3184  1124  980 S   0.0   0.0   0:00.00 SessionLeader
 4792 root       20   0 3200  1260 1108 S   0.0   0.0   0:00.01 Relay(4796)
 4796 srikar     20   0 6448  5628 3648 S   0.0   0.1   0:00.05 bash
 6200 srikar     20   0 9304  5760 3508 R   0.0   0.1   0:00.02 top
```

```
imvarls@LAPTOP-VQ4D70L7:~$ ps
  PID TTY          TIME CMD
   778 pts/0        00:00:00 bash
  1114 pts/0        00:00:00 ps
imvarls@LAPTOP-VQ4D70L7:~$ |
```

```
srikar@DESKTOP-ONIMS2G: ~
top - 05:01:26 up 1:24, 1 user, load average: 0.12, 0.12, 0.06
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 99.6 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7801.6 total, 7238.7 free, 528.6 used, 184.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7273.1 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
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    7 root        20   0 3196  2132 2008 S   0.0   0.0   0:00.00 init
   53 root       19  -1 50460 18584 17436 S   0.0   0.2   0:01.98 systemd-journal
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```

